Thesis Defense Preparation

Potential Questions & Answer Strategies

LBEADS-NET: Algorithm Unrolling for Learned Baseline Estimation and Denoising in Chromatographic Signals

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This document contains **65 anticipated defense questions** organized into eleven categories, each with a suggested answer strategy and references to specific thesis sections, tables, and equations. Section 11 contains the **killer questions**—the hardest, most uncomfortable challenges. A final section catalogues known inconsistencies to fix.

Contents

1 Foundational / "Why This Problem" Questions

Q1

Why chromatographic baseline correction specifically? Why not apply algorithm unrolling to general signal denoising, where the impact would be broader?

Answer Strategy

Key points:

- Chromatography has a *domain-specific structure* that generic denoisers ignore: the signal decomposes into exactly three additive components—baseline ($\mathbf{b}$), peaks ($\mathbf{p}$), and noise ($\boldsymbol{\varepsilon}$)—with distinct spectral and sparsity priors (Sec. 2.1).
- BEADS already encodes those priors via asymmetric penalty and highpass filtering. Unrolling preserves the interpretable structure while letting each layer adapt.
- Regulatory context matters: pharmaceutical QC (USP ⟨621⟩) and metabolomics both require *auditable* preprocessing, which favours interpretable, model-based architectures over black-box nets.
- General denoising is well-served by DnCNN / U-Net variants; baseline estimation is an under-explored niche where model-based deep learning is a natural fit.

Thesis references: Sec. 1.1 (motivation), Sec. 2.1 (chromatographic model), Sec. 2.3.3 (BEADS formulation).

Q2

What is wrong with just using arPLS or airPLS? They are simpler, they do not need training data, and they are already widely adopted in chemometrics. Convince me that the added complexity of unrolling is worth it.

Answer Strategy

Key points:

- arPLS/airPLS are iteratively-reweighted penalized least-squares methods. They require manual selection of the smoothness parameter λ and the asymmetry weight p, which are signal-dependent (Sec. 2.3.1–2.3.2).
- They perform *only* baseline estimation—they do not simultaneously denoise. BEADS and LBEADS-NET jointly estimate baseline and denoise.
- The core contribution is not "better numbers than arPLS" but *demonstrating the unrolling paradigm* in this domain and exposing the fundamental ℓ_1 leakage limitation.
- Concede honestly: on well-tuned, non-dense-peak data, arPLS will often suffice. LBEADS-NET is interesting as a research vehicle for understanding leakage.

Thesis references: Table 2.1 (baseline methods comparison), Sec. 2.3.1–2.3.2 (arPLS, airPLS).

Q3

Why does baseline drift matter quantitatively? What error magnitudes are we talking about in real instruments, and how does that translate to, say, drug concentration accuracy?

Answer Strategy

Key points:

- Column bleed, solvent gradients, and detector drift produce baseline offsets of 1–10% of the tallest peak height in HPLC (Sec. 2.1.2).
- In pharmaceutical QC, area-percent quantitation is the norm. A 2% baseline offset can shift a peak area by 5–15%, which exceeds the typical $\pm 2\%$ USP acceptance criterion.
- Metabolomics (hundreds of analytes per run) amplifies the problem: even small per-peak errors compound across the profile.
- Acknowledge: we do not empirically quantify this in the thesis because we work with synthetic data. Cite the chromatography literature for representative magnitudes.

Thesis references: Sec. 2.1.2 (baseline sources), Sec. 1.1 (quantitative motivation).

Q4

Who actually uses this in practice—pharmaceutical QC, metabolomics, environmental labs? Have you talked to anyone who processes real chromatograms and asked whether they need this?

Answer Strategy

Key points:

- Be honest: this is a methods thesis, not a deployment study. No practitioner interviews were conducted.
- Frame the contribution as *foundational analysis*: we unrolled BEADS, exposed the ℓ_1 leakage limit, and documented the failure modes. That knowledge benefits future applied work.
- Note that Professor Selesnick co-authored the original BEADS algorithm, so this work sits in a signal-processing research lineage rather than a product-development one.
- Mention potential users: Agilent OpenLab, Waters Empower, Shimadzu LabSolutions all rely on classical baseline algorithms that could benefit from learned parameter tuning.

Thesis references: Sec. 1.1, Sec. 6 (future work).

Q5

Why not just use a black-box neural network—a U-Net or a 1-D ResNet? They clearly generalise better and do not suffer from ℓ_1 leakage. What does unrolling buy you over a purely learned architecture?

Answer Strategy

Key points:

- **Parameter efficiency:** LBEADS-NET has **100 learnable scalars** (5×20 layers). A comparable 1-D U-Net would have 10^5–10^6 parameters, requiring far more training data than our 200 synthetic signals (Sec. 3.2, Sec. 4.1).
- **Interpretability:** every learned parameter has a physical meaning—λ_0 controls fidelity, λ_1/λ_2 control sparsity, r controls penalty shape, η is the step size. This matters for regulatory contexts.
- **Inductive bias:** the unrolled architecture *guarantees* the three-component decomposition structure. A black-box net must learn this from data alone.
- **Honest concession:** black-box nets would likely achieve lower reconstruction error given enough data. The trade-off is interpretability and data efficiency versus raw performance.

Thesis references: Sec. 2.4 (algorithm unrolling theory), Sec. 3.2 (architecture), Table 3.2 (parameter count).

2 Algorithm Unrolling Theory

Q6

What convergence guarantees does LBEADS-NET inherit from the classical BEADS algorithm? Can you prove that the learned iterates converge to anything meaningful?

Answer Strategy

Key points (this is a trap question—acknowledge it):

- Classical BEADS is derived via majorisation-minimisation (MM) and converges to the global optimum of its convex cost function (Sec. 2.3.3, Eq. 2.24).
- **Untied parameters break MM convergence.** Each layer minimises a *different* surrogate, so the sequence of iterates does not descend a single objective.
- This is a known trade-off in all algorithm unrolling: we sacrifice convergence guarantees for task-specific performance via end-to-end training (Sec. 2.4.3).
- Cite Chen & Pock (2017) and Monga et al. (2021): unrolled networks should be understood as *feed-forward architectures with structured layers*, not as truncated optimisers.
- The network is still well-posed in that training converges and test loss is stable—but "convergence of the iterates" in the MM sense does not apply.

Thesis references: Sec. 2.4.3 (unrolling trade-offs), Sec. 3.3 (ISTA reformulation).

Q7

Why ISTA over FISTA? You lose the $O(1/k^2)$ momentum acceleration for free. Was this a deliberate choice or just simpler to implement?

Answer Strategy

Key points:

- The primary reason was **avoiding gradient vanishing through $K \times K_{\mathbf{inner}}$ depth**. Classical BEADS uses a conjugate-gradient inner loop; unrolling both outer MM and inner CG iterations creates excessive depth (Sec. 3.3).
- ISTA replaces the inner CG loop entirely: each unrolled layer is a single proximal-gradient step, giving depth K instead of $K \times K_{\text{inner}}$.
- FISTA's momentum term introduces a non-monotone trajectory that complicates backpropagation through the unrolled graph (the Nesterov correction depends on the *previous two* iterates, increasing memory footprint).
- In the unrolled setting with untied parameters, the step-size $\eta^{(k)}$ is learned per layer and effectively subsumes much of the acceleration benefit.
- Concede: FISTA layers are a natural extension for future work.

Thesis references: Sec. 3.3 (ISTA reformulation), Sec. 2.4.2 (LISTA and ISTA-based unrolling).

Q8

How does your approach differ from LISTA fundamentally? LISTA learns entire weight matrices, whereas you learn only scalars. Is LBEADS-NET actually "unrolling" or just "parameterised iteration"?

Answer Strategy

Key points:

- LISTA (Gregor & LeCun, 2010) unrolls ISTA for sparse coding: it learns the dictionary $\mathbf{W}$ and the Gram matrix $\mathbf{S}$ as free matrices, so each layer is a full affine transform (Sec. 2.4.2).
- LBEADS-NET retains the *fixed structural operators* ($\mathbf{H}$ highpass filter, $\mathbf{D}$ difference matrices) and only untie the *scalar regularisation weights* across layers—5 scalars per layer.
- This is a design choice for interpretability and parameter efficiency, not a limitation. The structural operators encode domain knowledge (filter design, finite differences) that should *not* be made freely learnable with only 200 training signals.
- It is still correctly termed "unrolling": each layer corresponds to one iteration of a known algorithm with layer-specific parameters. The Monga et al. (2021) taxonomy calls this "algorithm unrolling with untied hyperparameters."

Thesis references: Sec. 2.4.2 (LISTA), Sec. 3.2 (LBEADS-NET architecture), Table 3.2.

Q9

What happens if you tie all parameters across layers—that is, force $\lambda_0^{(k)} = \lambda_0$ for all k? Does it reduce to classical BEADS? Have you tried it?

Answer Strategy

Key points:

- With tied parameters, each layer applies the *same* proximal-gradient step, so the unrolled network becomes K iterations of ISTA applied to the BEADS cost function with *learned* but *shared* regularisation weights.
- It does **not** exactly reduce to classical BEADS because (a) BEADS uses MM with CG inner solve, not ISTA, and (b) the $K = 20$ iteration budget may not suffice for convergence of the tied-parameter ISTA.
- The tied-parameter variant is an important ablation that isolates the benefit of *per-layer specialisation.*
- Concede: this ablation is listed in Table 5.3 but has not been run yet. It is planned before the final submission.
- The emergent parameter specialisation (Sec. 5.2.3) suggests that untying is beneficial: early layers learn coarse separation, later layers learn aggressive refinement.

Thesis references: Sec. 3.2 (architecture), Sec. 5.2.3 (parameter analysis), Table 5.3 (ablation, currently empty).

Q10

The BEADS cost function is convex. Does unrolling preserve convexity of the end-to-end mapping? If not, what are the optimisation implications for training?

Answer Strategy

Key points:

- **No.** The end-to-end mapping $\mathbf{y} \mapsto (\hat{\mathbf{b}}, \hat{\mathbf{p}})$ is a composition of 20 layers, each involving soft-thresholding (non-linear) and matrix multiplications parameterised by learned scalars. The composition is *not* convex in the parameters $\{\lambda_0^{(k)}, \ldots, \eta^{(k)}\}$.
- The *training loss landscape* is therefore non-convex, which is standard for neural networks.
- However, each *individual layer* is a proximal operator applied to a convex penalty, which provides local regularity (Lipschitz continuity of the proximal map).
- Practically, the 3-stage curriculum (Sec. 3.8) and progressive loss introduction help navigate the non-convex landscape by providing a good initialisation path.

Thesis references: Sec. 2.3.3 (BEADS convexity), Sec. 3.3 (ISTA layers), Sec. 3.8 (curriculum training).

3 Architecture & Design Choices

Q11

Why K=20 layers? Your ablation table (Table 5.3) is completely empty. How do you justify this choice without any empirical evidence?

Answer Strategy

Key points:

- Honest answer: K=20 was chosen heuristically based on (a) the classical BEADS default of 20–30 outer iterations and (b) GPU memory constraints with N=4096 and dense matrices.
- The parameter analysis in Sec. 5.2.3 was conducted with K=6 and N=1024, which is not directly comparable.
- Acknowledge that the empty ablation table is a gap. State that $K \in \{6, 10, 15, 20, 30\}$ should be swept.
- Qualitative argument: BEADS typically converges in ∼15–20 iterations, so the unrolled variant should need *at most* that many layers (and likely fewer, since parameters are untied).
- Cite the LISTA literature: Borgerding et al. (2017) show that unrolled ISTA with 15–20 layers matches 100+ iterations of classical ISTA.

Thesis references: Table 5.3 (empty ablation), Sec. 5.2.3 (6-layer analysis), Sec. 3.2.

Q12

Your implementation uses dense $N \times N$ matrices with $O(N^2)$ memory. For N=4096 in float64, each matrix is 128 MB. With 20 layers, you are storing gigabytes of intermediate buffers. Why not exploit the banded structure of $\mathbf{H}$ and $\mathbf{D}$?

Answer Strategy

Key points:

- You are correct—this is the single largest engineering limitation of the current implementation.
- The highpass filter $\mathbf{H}$ is a banded Toeplitz matrix and $\mathbf{D}$ (first/second difference) is banded with bandwidth 1–2. Products $\mathbf{H}^T\mathbf{H}$ and $\mathbf{D}^T\mathbf{D}$ are also banded.
- The reason for dense matrices: the filter $\mathbf{H}$ is constructed via `scipy.signal` and converted to a dense NumPy array before wrapping in a PyTorch tensor. Implementing banded-matrix operations natively in PyTorch would require custom CUDA kernels or use of `torch.sparse`.
- This is an engineering priority for future work. Banded storage would reduce memory from $O(N^2)$ to $O(N \cdot b)$ where b is the bandwidth ($b \ll N$), enabling N=65536+ signals.
- Practically, the current implementation fits on a single GPU for N=4096 only because of small batch sizes.

Thesis references: Sec. 3.4 (filter construction), Sec. 6 (future work).

Q13

You say f_c (the highpass cutoff frequency) is not learnable because of the "SciPy–PyTorch autograd boundary." But why not implement the entire filter construction in PyTorch? The Butterworth poles-and-zeros are closed-form.

Answer Strategy

Key points:

- The current implementation calls `scipy.signal.butter()` and `scipy.signal.tf2zpk()`, which are not differentiable through PyTorch's autograd (Sec. 3.4).
- In principle, the bilinear-transform design of a Butterworth filter *can* be written as closed-form operations on f_c (pole placement on the unit circle), and these operations are differentiable.
- The difficulty is constructing the *banded matrix* $\mathbf{H}(f_c)$ from those poles in a way that is compatible with efficient batched backpropagation.
- A more practical route: parameterise the filter as a learnable FIR filter of fixed order and initialise it from the Butterworth design. This avoids the IIR stability issues.
- This is listed as future work (Sec. 6) and would add one more learnable parameter per layer (101 total scalars).

Thesis references: Sec. 3.4 (filter construction), Eq. 3.5–3.6, Sec. 6 (future work).

Q14

What is the effective receptive field of LBEADS-NET? Each layer operates on the full signal via dense matrix multiplication, so is the receptive field trivially the entire signal?

Answer Strategy

Key points:

- Yes—formally, every output sample depends on every input sample after a single layer, because $\mathbf{H}$ is an IIR filter realised as a dense matrix (Sec. 3.4).
- However, the *effective* receptive field (in the Luo et al. 2016 sense) is narrower because the energy of $\mathbf{H}$ decays exponentially away from the diagonal.
- The practical receptive field is governed by the filter order and f_c=0.006: the $-3\,\text{dB}$ point at 0.6% of Nyquist corresponds to a spatial scale of $\sim$167 samples.
- For baseline estimation, a large receptive field is *desirable*: the baseline is a slowly-varying component that depends on global signal context.
- For peak estimation, local structure matters more, which is why the sparsity penalties (ℓ_1, TV) act element-wise or on adjacent differences.

Thesis references: Sec. 3.4 (filter design), Sec. 2.3.3 (BEADS formulation).

Q15

Why float64 instead of float32? Is this a fundamental numerical requirement of the algorithm, or is it just an engineering choice that doubles your memory usage?

Answer Strategy

Key points:

- It is partially fundamental: the highpass filter $\mathbf{H}$ at f_c=0.006 has eigenvalues spanning several orders of magnitude. In float32, the matrix $\mathbf{H}^T\mathbf{H} + \lambda_0\mathbf{I}$ can become poorly conditioned and produce NaN gradients during backpropagation.
- Empirically, early experiments in float32 showed training instability (gradient overflow in layers 15–20).
- However, with banded solvers or mixed-precision strategies (float64 for the filter, float32 for the rest), it should be possible to reduce memory. This was not explored.
- The ISTA reformulation partially mitigates the conditioning issue (no matrix inverse required), so float32 may be viable with careful step-size initialisation.

Thesis references: Sec. 3.4 (numerical considerations), Sec. 4.2 (training configuration).

Q16

You mention using softplus for non-negativity constraints in v7 but not in v5. Softplus versus ReLU—what is the difference in this context, and why does it matter?

Answer Strategy

Key points:

- The regularisation parameters $\lambda_0, \lambda_1, \lambda_2, r, \eta$ must be strictly positive. A reparameterisation is needed: the network stores unconstrained $\theta^{(k)}$ and maps to $\lambda^{(k)} = g(\theta^{(k)})$.
- **ReLU:** $g(\theta) = \max(0, \theta)$. Has zero gradient for $\theta < 0$ ("dead parameter" problem) and is non-differentiable at 0.
- **Softplus:** $g(\theta) = \log(1 + e^{\theta})$. Smooth, strictly positive, non-zero gradient everywhere.
- In v7, softplus was adopted after observing that some parameters collapsed to zero under ReLU and could not recover (gradient was exactly zero).
- Softplus provides a mild bias toward small positive values rather than exact zeros, which is physically appropriate—the regularisation weights should not be zero.

Thesis references: Sec. 3.6 (parameter constraints), Sec. 3.7.1 (v7 modifications).

4 Loss Function Deep Dives

Q17

You have 12 loss terms with 12 hyperparameters (the weights). Classical BEADS has 6 parameters. Haven't you just replaced one tuning problem with a larger one? How is this progress?

Answer Strategy

Key points:

- Fair criticism—acknowledge it directly.
- The critical difference: BEADS's 6 parameters must be tuned *per signal* (or per instrument/method), whereas the 12 loss weights are set *once during training* and the network then adapts its 100 per-layer parameters to each new signal at inference time.
- The loss weights are meta-parameters that define the *training objective*, not the inference-time model. At test time, LBEADS-NET has zero user-tunable parameters.
- Concede: the proliferation of loss terms (especially the 12-term v7 variant) is itself a finding—Experiment 1.5 shows that the full 12-term loss *hurts* performance due to competing gradients (Sec. 5.2).
- The thesis's *negative result*—that more loss terms do not help—is itself a contribution.

Thesis references: Table 3.3 (loss terms), Sec. 5.2.1 (Exp. 1.5 results), Sec. 3.7.

Q18

The orthogonality loss $\mathcal{L}_{orth}$ uses `.detach()`'d ground-truth-derived weights. Isn't this a form of label leakage during training? You are giving the network access to the answer.

Answer Strategy

Key points:

- Clarify what happens: the ground-truth peak signal is used to construct a *weighting mask* that emphasises peak regions, and this mask is detached (no gradient flows through it). The loss penalises $\langle \hat{\mathbf{b}}, \mathbf{w} \rangle$ where $\mathbf{w}$ is the detached mask.
- This is **not** label leakage in the test-time sense: the mask is used only during training (supervised learning always uses ground truth). At inference, no ground truth is available or used.
- The `.detach()` prevents the gradient from flowing back through the mask construction, which would create a circular dependency.
- Analogy: it is similar to importance sampling in RL, where the sampling distribution depends on the target but gradients are blocked through the weights.
- However, this does make the loss non-differentiable with respect to the mask shape, so the network cannot learn to "game" the mask.

Thesis references: Sec. 3.7 (loss function definition), Eq. 3.18 (orthogonality loss).

Q19

Your asymmetric baseline loss uses α=0.9 (a 9:1 ratio penalising over-estimation versus under-estimation). How sensitive is the model to this choice? Did you try α=0.7 or α=0.95?

Answer Strategy

Key points:

- The intuition for α=0.9: baseline over-estimation absorbs peak energy (leakage), which is harder to correct downstream than a small under-estimation. So we penalise over-estimation 9$\times$ more heavily.
- Concede: a sensitivity sweep over α was not performed. This is a gap.
- Argue that the exact value matters less than the *qualitative asymmetry*: any $\alpha > 0.5$ biases the baseline downward, and the effect saturates.
- Very high α (e.g., 0.99) would force the baseline to zero everywhere, destroying the baseline estimate. Very low α (e.g., 0.6) provides weak anti-leakage pressure.
- A proper sweep (Table 5.4) is listed but not yet run.

Thesis references: Sec. 3.7 (asymmetric loss), Table 5.4 (sensitivity ablation, currently empty).

Q20

The envelope constraint uses a soft-minimum with temperature τ=0.1 and a window size of 51. How sensitive are the results to these choices? Why 51?

Answer Strategy

Key points:

- Window size 51 corresponds to roughly the half-width of a typical synthetic peak ($\sigma \approx$ 20–30 samples), so the envelope captures the local minimum in the valley between adjacent peaks.
- Temperature τ=0.1 makes the soft-min approximate a hard min: $\mathrm{softmin}_\tau(\mathbf{x}) \to \min(\mathbf{x})$ as $\tau \to 0$. At τ=0.1, the approximation is tight while remaining differentiable.
- Sensitivity: smaller windows ($\sim$21) would miss wide peaks; larger windows ($\sim$101) would create a baseline that cannot follow moderate-frequency drift.
- This was not swept systematically. The values were set once based on the known peak widths in the synthetic data.
- On real data with unknown peak widths, these would need adaptive selection—a limitation.

Thesis references: Sec. 3.7 (envelope constraint definition).

Q21

You describe the ℓ_1 and TV penalties as both *the cause of baseline leakage* and *necessary for sparsity in non-peak regions. This seems like a fundamental contradiction. How do you resolve it?*

Answer Strategy

Key points:

- This is the **central tension of the thesis**—do not try to resolve it cleanly, because it *is not resolved*.
- ℓ_1 promotes sparsity in $\mathbf{p}$, which is correct in regions with no peaks (sparse support). But in dense-peak regions, ℓ_1 biases peak amplitudes toward zero (*soft-thresholding shrinkage bias*, well-known in compressed sensing).
- The energy removed from peaks has to go somewhere in the $\mathbf{y} = \mathbf{b} + \mathbf{p} + \boldsymbol{\varepsilon}$ decomposition—it leaks into $\hat{\mathbf{b}}$.
- The 12-term loss attempts to *mitigate* this with anti-leakage terms, but BLI plateaus at $\sim$0.59, confirming that loss engineering cannot overcome the architectural bias.
- **The honest conclusion**: ℓ_1-based unrolling is fundamentally limited for dense-peak signals. Future work should explore group sparsity, ℓ_0-like penalties, or hybrid architectures.

Thesis references: Sec. 3.7 (loss design rationale), Sec. 5.2.2 (BLI plateau analysis), Sec. 5.5 (leakage discussion).

Q22

You use Huber loss with δ=1.0 for reconstruction in the main experiments but mention δ=0.1 elsewhere. Which is it, and why Huber over MSE in the first place?

Answer Strategy

Key points:

- Huber loss is piecewise: quadratic for $|e| \leq \delta$, linear for $|e| > \delta$. It down-weights large residuals, providing robustness to outlier peaks.
- δ=1.0: the data is normalised to $[0, 1]$, so δ=1.0 means *almost all residuals fall in the quadratic regime*—$\text{Huber}_{1.0} \approx \text{MSE}$ for normalised data.
- δ=0.1: makes the linear regime much larger, acting more like MAE. This was used in the curriculum's Stage B (Sec. 3.8) to reduce sensitivity to large leakage residuals during the early anti-leakage phase.
- Acknowledge that the δ value is not consistently documented and should be clarified.
- MSE was not used because early experiments showed that squared large residuals from leakage regions dominated the gradient, destabilising training.

Thesis references: Sec. 3.7 (loss terms), Sec. 3.8 (curriculum training), Table 4.2 (training config).

5 Experimental Methodology

Q23

200 signals total, 40 for testing. How can you draw any *statistically meaningful conclusions from 40 test signals? What is your confidence interval on, say, peak correlation?*

Answer Strategy

Key points:

- Concede: 40 test signals is small. No confidence intervals or statistical tests are reported, which is a limitation.
- Mitigating factors: each signal contains 8–25 peaks, so the effective sample size for *per-peak* metrics is $\sim$600. Per-signal metrics are averaged over $\sim$40 diverse signals.
- The synthetic data generation is controlled, so there is no domain shift between train and test—only random variation in peak parameters.
- The small dataset is *by design*: it tests whether unrolling's inductive bias can compensate for data scarcity. A 100k-signal dataset would not test this hypothesis.
- For a proper statistical treatment, report mean $\pm$ std and bootstrap confidence intervals. This should be added before the final submission.

Thesis references: Sec. 4.1 (data generation), Sec. 5.1 (evaluation metrics), Table 5.1.

Q24

Your comparison with classical BEADS uses parameters designed for unnormalised data, applied to normalised data. Isn't this a deliberately unfair comparison that makes BEADS look worse than it is?

Answer Strategy

Key points:

- Partially valid criticism. The default BEADS parameters ($\lambda_0{=}1, \lambda_1{=}1, \lambda_2{=}1$) are recommended for unnormalised chromatograms with peak heights of 10^3–10^6. On $[0, 1]$-normalised data, these regularisation strengths are relatively too large.
- However: the comparison is intended to show what happens when a user applies BEADS "out of the box" without per-signal tuning. This is the common use case.
- A *fairer* comparison would grid-search BEADS parameters on a validation set. This was not done and should be acknowledged as a limitation.
- The main contribution is not "LBEADS-NET beats BEADS" but rather the analysis of leakage and the unrolling methodology.

Thesis references: Sec. 5.2 (comparison setup), Table 5.1 (results).

Q25

You survey arPLS, airPLS, SNIP, and several deep learning baselines in Chapter 2, but you never compare against any *of them experimentally. Why not?*

Answer Strategy

Key points:

- Concede: this is a significant omission.
- The reason is scope and time: implementing each baseline with proper hyperparameter tuning on the synthetic dataset would require substantial additional work.
- The thesis's primary goal is to unroll *BEADS specifically* (given the advisor's authorship of BEADS), not to conduct a broad benchmark.
- arPLS is the most important missing comparison because it is the most widely used method in practice. This should be added before the defense if possible.
- Deep learning baselines (1-D U-Net) would require architectural design and training, which is a separate study.
- Frame as future work.

Thesis references: Sec. 2.3 (method survey), Table 2.1 (comparison table), Sec. 6 (future work).

Q26

Your synthetic data uses bi-Gaussian peaks (Section 4.1.1), but your background chapter discusses the EMG (Exponentially Modified Gaussian) model in detail (Section 2.1.4). Why the mismatch? Does this limit the relevance of your results?

Answer Strategy

Key points:

- EMG is the standard physical model for chromatographic peaks (mass transfer kinetics cause exponential tailing). Bi-Gaussian is a simpler empirical model that captures asymmetry without the convolution integral.
- Bi-Gaussian was chosen for computational simplicity in synthetic data generation and because it has an analytic closed-form (no numerical convolution needed).
- The two models produce qualitatively similar peak shapes for moderate asymmetry. For heavy tailing, EMG produces longer tails that bi-Gaussian cannot capture.
- This mismatch is not explained in the thesis and should be. Add a brief justification in Sec. 4.1.1.
- The choice does limit relevance: results on bi-Gaussian peaks may not transfer to heavily tailed EMG peaks.

Thesis references: Sec. 2.1.4 (EMG model), Sec. 4.1.1 (bi-Gaussian generation).

Q27

Your synthetic data generation has many design choices: cluster probability 0.4, log-uniform amplitude distributions, specific σ ranges. How sensitive are your results to these choices? Have you tested on a different distribution?

Answer Strategy

Key points:

- These parameters control the "difficulty distribution" of the training set: cluster probability determines how often dense-peak regions occur (which is where leakage is worst).
- No sensitivity analysis on data generation parameters was performed. This is a limitation.
- Higher cluster probability would create more dense-peak signals, likely increasing average BLI. Lower cluster probability would create easier signals with better metrics.
- The log-uniform amplitude distribution ensures coverage of both small and large peaks, which is more realistic than uniform sampling.
- A robust model should be tested on out-of-distribution data (different peak width ranges, different noise levels). This was not done.

Thesis references: Sec. 4.1.1 (data generation parameters), Table 4.1 (generation config).

6 Results Interpretation

Q28

BLI is approximately 0.59 across all *configurations—from the simplest 4-term loss to the full 12-term v7 loss. Doesn't this mean your entire loss engineering effort was basically useless for addressing leakage?*

Answer Strategy

Key points:

- **Yes, essentially.** This is a key negative result and should be stated directly.
- The BLI plateau at ∼0.59 across all configurations is strong evidence that leakage is an **architectural** property of ℓ_1-based unrolling, not a loss-function problem.
- This is a *valuable finding*: it tells future researchers not to invest in loss engineering for this architecture. The fix must be structural (different penalty, different architecture).
- The loss terms *do* help other metrics (reconstruction error, peak shape fidelity), just not leakage.
- Frame this as: "the thesis *discovered* the leakage floor and *proved* that loss engineering cannot break through it."

Thesis references: Table 5.1 (BLI column), Sec. 5.2.2 (BLI analysis), Sec. 5.5 (leakage discussion).

Q29

Area error is 49–58% across all LBEADS-NET variants. In pharmaceutical QC, 1% accuracy is the requirement. 50% error is catastrophic. How can you call this a contribution?

Answer Strategy

Key points:

- Do not try to defend 50% area error as acceptable—it is not, and any committee member from chromatography would dismiss this immediately.
- Reframe: the contribution is *not* a production-ready tool. It is a **methodological study** that: (a) demonstrates algorithm unrolling for BEADS, (b) identifies and characterises the ℓ_1 leakage limit, (c) documents the failure of loss engineering to fix it.
- The area error is dominated by leakage in dense-peak regions. For isolated peaks, the per-peak area error is much lower (this should be reported separately if data is available).
- The hybrid pipeline (Sec. 3.9) improves area error by applying classical BEADS post-processing, but the fundamental limit remains.
- Honest conclusion: LBEADS-NET in its current form is not suitable for quantitative chromatography.

Thesis references: Table 5.1 (area error), Sec. 5.2 (quantitative results), Sec. 5.5.

Q30

Experiment 1.5 (the full v7 12-term loss) performs worse *than Experiment 1.3 (a simpler configuration). Your own results show that simpler is better. Why do you propose the complex version at all?*

Answer Strategy

Key points:

- This is a genuinely important finding. The 12-term loss creates **competing gradient directions**: the reconstruction loss pulls parameters one way while the anti-leakage losses pull them another way, resulting in a compromise that is worse on all metrics.
- This is an example of *over-regularisation*—adding too many constraints to a 100-parameter model.
- The simpler Experiment 1.3 succeeds because fewer loss terms give the optimiser a cleaner gradient signal.
- The thesis proposes v7 as a *research contribution* (novel loss formulations) even though v5/Exp. 1.3 is empirically better. The intellectual value is in understanding *why* the complex version fails.
- Recommendation: the thesis should more prominently highlight Exp. 1.3 as the best-performing configuration.

Thesis references: Table 5.1 (Exp. 1.3 vs 1.5), Sec. 5.2.1 (progressive results).

Q31

Peak correlation maxes at 0.793. What does this mean physically? That 79% of the peak shape is recovered? What is an acceptable value in practice?

Answer Strategy

Key points:

- Correlation measures *shape similarity* (cosine of the angle between the estimated and true peak vectors), not amplitude accuracy. A correlation of 0.793 means the estimated peaks preserve ∼79% of the shape structure.
- The remaining ∼21% discrepancy comes from (a) amplitude shrinkage due to ℓ_1 bias and (b) baseline leakage distorting the peak shape, especially at the base.
- In practice, $r > 0.95$ is expected for quantitative chromatography. 0.793 would not be acceptable.
- For qualitative analysis (peak detection, retention time identification), 0.793 may be sufficient since the peak *positions* are typically preserved even when amplitudes are distorted.
- The gap from 0.793 to 1.0 is another manifestation of the ℓ_1 leakage problem.

Thesis references: Sec. 5.1 (metric definitions), Table 5.1 (peak correlation).

Q32

The learned parameter analysis (Section 5.2.3) uses a 6-layer model with N=1024, but the main results use a 20-layer model with N=4096. Are these comparable? Why not analyse the parameters of the actual model you evaluate?

Answer Strategy

Key points:

- They are **not directly comparable**, and this should be acknowledged.
- The 6-layer/N=1024 model was used for the parameter analysis because (a) it is easier to visualise 6 layers and (b) it was an earlier experiment before the full 20-layer model was trained.
- The qualitative pattern (early layers learn coarse separation, later layers learn refinement) is expected to hold across depths, but the specific parameter values would differ.
- Ideally, the parameter trajectories of the 20-layer model should be plotted. This is a gap.
- The analysis still provides insight into the *specialisation phenomenon*, which is the main finding of that section, even if the numerical values are model-specific.

Thesis references: Sec. 5.2.3 (parameter analysis), Sec. 4.2 (model configurations).

7 Baseline Leakage Deep Dive

Q33

You claim leakage is a "fundamental limit of ℓ_1-based formulations." But classical BEADS with properly tuned parameters on unnormalised data presumably has lower leakage. Isn't the leakage just a parameter-tuning problem that you failed to solve?

Answer Strategy

Key points:

- The leakage mechanism is *not* a tuning artefact: it stems from the soft-thresholding operator $\text{soft}(x, \lambda) = \text{sign}(x)(|x| - \lambda)_+$, which *always* shrinks non-zero entries by exactly λ. This bias exists for any $\lambda > 0$.
- Classical BEADS on unnormalised data has the *same* mechanism, but the effect may be less visible because (a) peak amplitudes are 10^3–10^6 so the relative shrinkage λ/A is small, and (b) users tune λ to minimise visible artefacts.
- On normalised $[0, 1]$ data, the shrinkage λ/A is relatively larger, making leakage more prominent.
- The deeper point: in *dense-peak regions*, the sparsity assumption fails entirely (peaks overlap, the signal is not sparse), so ℓ_1 is the wrong prior.
- The leakage is thus both a tuning issue (partially) *and* a structural issue (fundamentally, in dense regions).

Thesis references: Sec. 5.5 (leakage analysis), Sec. 3.5 (proximal operator), Eq. 3.11.

Q34

You mention group sparsity as a potential solution to leakage. Can you explain concretely how it would help, and why you did not pursue it?

Answer Strategy

Key points:

- Group sparsity (e.g., group LASSO or $\ell_{2,1}$ penalty) penalises *groups* of adjacent coefficients jointly: $\sum_g \|\mathbf{p}_g\|_2$. Either the entire group is zero or non-zero—no element-wise shrinkage within active groups.
- This directly addresses the leakage mechanism: within an active peak group, amplitudes are *not* shrunk toward zero, preserving peak shape and area.
- Implementation challenge: defining the groups. Peak boundaries are unknown a priori, so the grouping must be adaptive or overlapping (e.g., sliding-window group LASSO).
- The proximal operator for group $\ell_{2,1}$ is a *block* soft-thresholding, which is straightforward to implement in PyTorch.
- Not pursued due to scope and time constraints. This is the highest-priority future work.

Thesis references: Sec. 5.5 (leakage discussion), Sec. 6 (future work).

Q35

Gharbi et al.'s U-HQ method uses a hybrid penalty that avoids ℓ_1 shrinkage bias. You cite it in the background. Why not adopt their approach instead of fighting the ℓ_1 bias with loss terms?

Answer Strategy

Key points:

- U-HQ (Gharbi et al.) uses a non-convex penalty that approximates ℓ_0: it does not shrink large coefficients, only eliminates small ones. This would indeed avoid leakage.
- The challenge: non-convex penalties lose the convergence guarantees of MM/ISTA. The proximal operator may not be unique, and unrolling a non-convergent algorithm is theoretically questionable.
- Adopting U-HQ would require redesigning the entire unrolling framework: different cost function, different proximal operator, different convergence analysis.
- This is the right direction for future work, but it is a separate thesis.
- The contribution here is establishing *why* the ℓ_1 version fails, which motivates the move to U-HQ-style penalties.

Thesis references: Sec. 2.3 (U-HQ discussion), Sec. 5.5 (leakage analysis), Sec. 6.

Q36

If baseline leakage is truly a fundamental limitation of the ℓ_1 formulation, then why build on BEADS at all? Isn't choosing a flawed base algorithm the core problem?

Answer Strategy

Key points:

- BEADS was chosen because (a) it is the advisor's algorithm, making this a natural extension of the group's research, and (b) it is one of the most widely used baseline estimation methods with a clear optimisation structure suitable for unrolling.
- BEADS is not "flawed"—it is a well-designed convex programme that works well for sparse-peak chromatograms. The limitation arises specifically in the dense-peak regime.
- The thesis demonstrates *exactly where* BEADS breaks down when unrolled. This is a contribution: understanding the limits of an existing method rigorously.
- The negative result is arguably more valuable than a modest positive improvement: it redirects future research toward structural solutions (group sparsity, non-convex penalties).
- "All models are wrong; some are useful." BEADS is useful; we now know precisely where it stops being useful.

Thesis references: Sec. 1.2 (thesis scope), Sec. 2.3.3 (BEADS), Sec. 5.5 (limitations).

8 Practical / "So What" Questions

Q37

Has LBEADS-NET been validated on even one *real chromatographic dataset with known ground truth?*

Answer Strategy

Key points:

- **No.** All experiments use synthetic data.
- The fundamental problem: real chromatograms do not have ground-truth decompositions ($\mathbf{b}$, $\mathbf{p}$, $\boldsymbol{\varepsilon}$ are not separately measurable).
- Semi-synthetic validation is possible: take a known peak standard (e.g., caffeine), subtract a fitted baseline manually, and use the result as "ground truth." This was not attempted.
- Public chromatographic datasets with ground truth are essentially non-existent. This is a known challenge in the field.
- Real-data validation is the most important next step (Sec. 6).

Thesis references: Sec. 4.1 (synthetic data only), Sec. 6 (future work).

Q38

What is the inference time of LBEADS-NET? If each layer requires a dense $N \times N$ matrix–vector multiply, it might be slower than classical BEADS, which is $O(N)$ per iteration with banded solvers.

Answer Strategy

Key points:

- Inference time is **not reported** in the thesis—this is a gap.
- Each of the K=20 layers performs $O(N^2)$ operations due to dense matrix multiplication. For N=4096, this is ∼20 × 16M = 320M flops.
- Classical BEADS with banded solvers is $O(N \cdot b \cdot K)$ where b is bandwidth (∼5–10), so ∼20 × 40K = 800K flops—**400× fewer**.
- On a GPU, the dense matmuls are parallelised, so wall-clock time may be comparable. But on CPU (the typical analytical lab environment), LBEADS-NET is significantly slower.
- With banded implementation, LBEADS-NET would also be $O(N)$ per layer, eliminating the gap.

Thesis references: Sec. 3.4 (implementation), Sec. 6 (future work on banded ops).

Q39

How would a chromatographer actually use LBEADS-NET? They do not have ground-truth baselines for training. Does the model generalise to unseen instrument configurations?

Answer Strategy

Key points:

- Usage scenario: train on synthetic data that covers the expected range of peak shapes, baseline curvatures, and noise levels; deploy on real data.
- This is the standard paradigm for model-based deep learning in domains without ground truth (e.g., MRI reconstruction).
- Generalisation to unseen instruments depends on whether the synthetic data distribution covers the real data distribution. This is not tested.
- Domain adaptation (fine-tuning on a small set of manually corrected real chromatograms) is a natural extension.
- Practically, the user would run LBEADS-NET as a preprocessing step in their chromatography data system (CDS) software, replacing the built-in baseline correction module.

Thesis references: Sec. 4.1 (synthetic data rationale), Sec. 6 (deployment discussion).

Q40

Your hybrid pipeline adds classical BEADS as a post-processing step. If you still need classical BEADS at the end, what did unrolling actually buy you?

Answer Strategy

Key points:

- The hybrid pipeline uses LBEADS-NET to provide an *initial decomposition* that is closer to the true solution than the default initialisation ($\mathbf{b}_0 = 0$). Classical BEADS then refines from this warm start.
- The benefit: BEADS converges in fewer iterations from a good initialisation, and the learned parameters provide signal-adaptive regularisation weights that BEADS alone cannot.
- Concede: if classical BEADS with proper tuning achieves similar results, the unrolling step adds complexity without clear benefit.
- The honest framing: the hybrid pipeline is a *fallback strategy* when LBEADS-NET alone is insufficient, not the primary contribution.
- The primary contribution is the understanding of leakage, not the pipeline itself.

Thesis references: Sec. 3.9 (hybrid pipeline), Sec. 5.2 (hybrid results).

9 Writing & Completeness

Q41

Your abstract is Lorem Ipsum. Your conclusions chapter is Lorem Ipsum. Your ablation tables have all dashes. Several sections are marked as placeholders. Is this thesis actually complete?

Answer Strategy

Key points:

- **Do not get defensive.** Acknowledge that these sections are incomplete and will be finalised before the official submission deadline.
- Emphasise that the *technical content* (Chapters 2–5) is complete and the defence focuses on the technical contribution.
- Provide a timeline: abstract and conclusions will be written after the defence, incorporating committee feedback.
- For the ablation tables: state which experiments are planned and the expected timeline.
- This is a common situation for defences that occur before the final manuscript submission.

Thesis references: Abstract, Chapter 6 (conclusions), Tables 5.3–5.4.

Q42

Several cross-references in your document show "Chapter ??" (pages 43, 56, 68). Have you compiled and checked the final PDF?

Answer Strategy

Key points:

- Acknowledge the broken references. These are caused by missing `\label{}` tags or mismatched `\ref{}` keys.
- State that a full compilation pass with `latexmk` (multiple runs to resolve references) will be done before submission.
- This is a mechanical fix, not a content issue.
- Have a plan: grep for "??" in the compiled PDF, trace each to the source `\ref{}` command, and fix.

Thesis references: Throughout document (cross-references).

Q43

You say "twelve-term composite loss" in Section 3.7 and Table 3.3, but your Introduction says "11-term." Which is correct?

Answer Strategy

Key points:

- The correct count is **twelve**. Section 3.7 and Table 3.3 enumerate all twelve terms explicitly.
- The Introduction's "11-term" is a typo from an earlier version of the loss function before the twelfth term (likely the envelope constraint) was added.
- This will be corrected in the Introduction before submission.
- Having the specific section and table to point to (Sec. 3.7, Table 3.3) provides a definitive answer.

Thesis references: Sec. 1.2 (Introduction, "11-term"), Sec. 3.7 ("twelve-term"), Table 3.3.

10 Future Work & Big Picture

Q44

If you had 6 more months, what would you do first and why?

Answer Strategy

Key points (prioritised):

1. **Group sparsity penalty** (2 months): replace ℓ_1 with $\ell_{2,1}$ or overlapping group LASSO. This directly addresses the leakage ceiling and is the highest-impact change.
2. **Banded matrix implementation** (1 month): exploit the sparsity structure of $\mathbf{H}$ and $\mathbf{D}$ to reduce memory from $O(N^2)$ to $O(N)$ and enable N=65536+ signals.
3. **Real data validation** (1 month): obtain chromatographic data from a collaborating lab, create semi-synthetic ground truth, and evaluate.
4. **arPLS/airPLS baseline comparison** (2 weeks): implement and tune arPLS on the existing synthetic dataset.
5. **Learnable f_c** (2 weeks): parameterise the highpass filter as a learnable FIR filter initialised from the Butterworth design.
6. **Complete ablations** (2 weeks): fill Tables 5.3 and 5.4.

Thesis references: Sec. 6 (future work), Sec. 5.5 (leakage analysis).

Q45

Could the LBEADS-NET approach generalise to other signal decomposition problems—Raman spectroscopy, NMR, mass spectrometry?

Answer Strategy

Key points:

- **Yes, in principle.** Any problem where the signal decomposes as $\mathbf{y} = \mathbf{b} + \mathbf{p} + \boldsymbol{\varepsilon}$ with a smooth baseline and sparse peaks is a candidate.
- **Raman spectroscopy**: fluorescence background is smooth and broadband, peaks are narrow Lorentzian—very similar structure. BEADS-like methods are already used (Zhang et al., 2010).
- **NMR**: baseline distortion from phase errors and receiver artefacts. The peak model is Lorentzian with known analytic form.
- **Mass spectrometry**: chemical noise creates a baseline, but the signal is often discrete (centroid data), which changes the problem structure.
- The unrolling framework is modular: replace the peak prior (Lorentzian instead of bi-Gaussian) and the filter design (adjust f_c for the spectral content), and retrain.
- The ℓ_1 leakage limitation would apply equally in all these domains—the proposed group sparsity fix would also be needed.

Thesis references: Sec. 2.1 (signal model), Sec. 6 (future work).

Q46

What would it take to make f_c learnable? Sketch the approach at the whiteboard.

Answer Strategy

Key points (sketch for the whiteboard):

1. **Current**: f_c enters through `scipy.signal.butter(d, f_c)` $\rightarrow$ IIR coefficients $\rightarrow$ dense matrix $\mathbf{H}$. This pipeline is not differentiable.
2. **Approach 1—Differentiable IIR:** Place Butterworth poles at $p_k = e^{j(\pi/2 \pm \Delta\theta_k)}$ where $\Delta\theta_k = g(f_c)$. Construct the transfer function $H(z) = \prod_k (1 - p_k z^{-1})$ in PyTorch. Apply via `torch.linalg.solve_triangular` (causal IIR). Gradients flow through $f_c \rightarrow p_k \rightarrow H(z) \rightarrow \mathbf{Hy}$.
3. **Approach 2—Learnable FIR:** Design an FIR filter of order M initialised from the Butterworth impulse response truncated at M taps. Make the tap weights $h_0, \ldots, h_M$ learnable. Apply via `torch.nn.functional.conv1d`. This is simpler but loses the sharp IIR rolloff.
4. **Approach 3—Frequency-domain:** Multiply $\mathbf{Y}(f)$ by a learnable frequency response $|\hat{H}(f)|$ parameterised by f_c and filter order. Invert via IFFT. Fully differentiable but introduces circular convolution artefacts.
5. **Recommended**: Approach 2 (FIR) for simplicity and robustness, with $M \sim 64$ taps.

Thesis references: Sec. 3.4 (current filter construction), Sec. 6 (future work).

11 Killer Questions — The Ones That Hurt

Warning

These are the questions designed to make you uncomfortable. They target circular reasoning, contribution validity, mathematical rigour, and the "so what" factor. If you can handle these, you can handle anything.

Q47

Strip away the loss engineering that your own results show does not work. What remains? You unrolled BEADS into ISTA layers with scalar parameters—that is just LISTA applied to a different cost function. Where, exactly, is the novelty?

Answer Strategy

Key points:

- Acknowledge the similarity to LISTA, but clarify the differences: (a) LISTA learns full matrices $\mathbf{W}_e, \mathbf{S}$; LBEADS-NET fixes the operators and learns only scalars. (b) LISTA targets sparse coding; LBEADS-NET targets a *three-component* signal decomposition with frequency-domain separation (baseline + peaks + noise).
- The novelty is application-specific: *no prior work has unrolled the BEADS algorithm.* The ISTA reformulation of the MM update (replacing CG inner solve with proximal gradient) is a design contribution.
- The identification and systematic characterisation of baseline leakage as a fundamental ℓ_1 limitation in this architecture is novel.
- The 12-term anti-leakage loss design (even though it fails to break the BLI floor) and the curriculum strategy are original engineering contributions.
- Concede: the theoretical novelty is modest. This is an *applied* thesis.

Thesis references: Sec. 2.3.2 (LISTA), Sec. 3.2–3.3 (ISTA reformulation), Sec. 1.2 (contributions).

Q48

You train on synthetic data and test on synthetic data generated by the same *code with the* same *distributions. This is a closed loop. You have proven nothing about real chromatography. How do you respond?*

Answer Strategy

Key points:

- This is correct, and it is a fundamental limitation.
- The synthetic data was calibrated against 8 real chromatograms (Sec. 4.1)—peak heights 10–2600+, widths FWHM 3–370+, 29–58 peaks per signal—but the test set is still synthetic.
- The thesis contribution is *algorithmic*: demonstrating unrolling and characterising leakage. The synthetic evaluation framework controls for confounding variables that real data would introduce.
- All supervised deep learning for baseline correction (Kensert et al., Chen et al., Han et al.) uses synthetic training data—this is standard practice because ground truth does not exist for real chromatograms.
- The honest response: real-data validation is the single most important missing piece and the top priority for future work.

Thesis references: Sec. 4.1 (data generation), Sec. 5.6 (real chromatogram discussion).

Q49

Your standard deviations are enormous. Peak MSE is $9.30 \pm 4.7 \times 10^{-3}$*. That means some signals have MSE of* 4.6×10^{-3} *and others have* 14.0×10^{-3}*—a* $3\times$ *range. Is your model actually learning anything consistent, or is it just averaging over easy and hard cases?*

Answer Strategy

Key points:

- The high variance is expected: the test set deliberately includes signals with very different peak densities (20–55 peaks), baseline shapes (spline, exponential, mixed), and noise levels ($\sigma_w \in [2, 6]$).
- A signal with 20 well-separated peaks and low noise is genuinely *easier* than one with 55 overlapping peaks and high noise. The variance reflects problem difficulty, not model inconsistency.
- To verify: stratify results by peak density (sparse vs dense) and report separately. This was not done but should be.
- The *relative* improvement over classical BEADS is consistent across signals (the std of the improvement would be more informative than the std of the raw metric).
- Concede: reporting only mean ± std is insufficient. Median, IQR, and per-signal scatter plots would be more informative.

Thesis references: Table 5.1 (standard deviations), Sec. 4.1 (data diversity).

Q50

You say LBEADS-NET has "only 100 learnable parameters." But the dense matrices $\mathbf{A}$*,* $\mathbf{B}$*, and* $\mathbf{L}$ *are each* $N \times N = 4096^2 \approx 16.7$ *million entries. These are frozen, not absent. The model capacity is dominated by these fixed operators. Isn't "100 parameters" misleading?*

Answer Strategy

Key points:

- This is a valid nuance. The *learnable* parameter count is 100 scalars. The *total model capacity* includes the fixed filter matrices, which encode substantial prior knowledge.
- However, the fixed matrices are determined entirely by two hyperparameters (f_c and d) and the signal length N. They are not "learned from data"—they are analytically derived from the Butterworth filter design.
- The analogy: a CNN with a fixed Gabor filter bank and 100 learnable classification weights also has "100 learnable parameters" even though the Gabor filters encode structure.
- The "100 parameters" claim is about *what backpropagation can adjust*, not about the total information content of the model. This distinction should be made clearer in the thesis.
- The correct framing: "100 learnable parameters, with fixed structural operators derived from the BEADS formulation."

Thesis references: Sec. 3.4 (architecture), Table 3.1 (parameters per layer).

Q51

Why not just use Bayesian optimisation to tune the 6 classical BEADS parameters on a validation set? That would be simpler than unrolling, fully interpretable, and might work just as well.

Answer Strategy

Key points:

- Bayesian optimisation (BO) over $\{f_c, d, r, \lambda_0, \lambda_1, \lambda_2\}$ on a validation set is a perfectly reasonable baseline. It was not compared against, and it should have been.
- Key difference: BO finds a *single global* parameter set optimised for the validation distribution. LBEADS-NET learns *per-layer* parameters that specialise across depth (Sec. 5.5)—this is fundamentally richer.
- BO-tuned BEADS would still suffer from the same ℓ_1 leakage in dense-peak regions (the leakage is structural, not a tuning issue). But BO would likely achieve better results than the "unfair" default-parameter comparison.
- Concede: BO-tuned BEADS is the missing comparison that would isolate the benefit of *unrolling* from the benefit of *better parameters*.
- This is arguably the most important missing experiment.

Thesis references: Sec. 4.5 (comparison methods), Sec. 5.1 (results).

Q52

Your three-stage curriculum training is just a manually designed training schedule. You replaced BEADS's 6 hand-tuned parameters with 12 hand-tuned loss weights plus *a hand-designed 3-stage curriculum with hand-picked epoch budgets. How is this reducing human design effort?*

Answer Strategy

Key points:

- This is a fair and damaging critique. Acknowledge it.
- The key distinction: the manual design effort is a *one-time cost* during model development. Once trained, the model requires *zero* user input at inference time. Classical BEADS requires per-signal tuning at every use.
- The curriculum and loss weights are meta-level design decisions analogous to choosing a network architecture in deep learning—they are set by the model developer, not the end user.
- Concede: the total design effort for LBEADS-NET (12 loss weights, 3 curriculum stages, epoch budgets, etc.) is *greater* than tuning BEADS for a single signal. The payoff is generalisation across signals.
- Automated hyperparameter optimisation (e.g., learning the loss weights via meta-learning) would strengthen this claim and is a natural extension.

Thesis references: Sec. 3.6 (curriculum), Table 3.2 (stage configuration), Sec. 3.7 (loss weights).

Q53

Gharbi et al. already showed in 2024 that ℓ_1-based unrolled networks exhibit shrinkage bias in chromatographic signal recovery. You cite their work. Your "fundamental insight" about baseline leakage is a confirmation of their finding, not a new discovery. What do you actually add beyond their result?

Answer Strategy

Key points:

- Gharbi et al. (2024) observed that ℓ_1-based unrolled methods (U-PD, U-ISTA) produce biased peak intensities compared to their U-HQ hybrid penalty. This is indeed the same underlying phenomenon.
- What LBEADS-NET adds: (a) **Quantification**: the BLI and PBER metrics provide the first *numerical characterisation* of the leakage magnitude (BLI ≈ 0.59). (b) **Systematic mitigation attempt**: 12 anti-leakage loss terms, progressive experiments, and the proof that loss engineering cannot break the floor. (c) **Different algorithm**: BEADS (MM-based) vs Gharbi's PD/ISTA/HQ formulations. (d) **Curriculum training**: novel for this domain.
- Concede: the conceptual finding is not new. The contribution is the *depth of analysis* and the specific characterisation for the BEADS architecture.
- Frame as: "We independently confirmed and quantified a phenomenon that Gharbi et al. identified qualitatively."

Thesis references: Sec. 2.3.5 (Gharbi et al.), Sec. 5.1.2 (BLI analysis), Sec. 1.3 (contributions).

Q54

At the end of the day: your best model has 50% area error, 0.79 peak correlation, and 0.59 BLI. A practising chromatographer would never deploy this. Your loss engineering does not fix leakage. Your ablation tables are empty. What exactly is the contribution to knowledge?

Answer Strategy

Key points (this is THE critical question—have this answer cold):

- **Contribution 1**: First unrolling of the BEADS algorithm. Demonstrated that BEADS can be reformulated as ISTA layers, trained end-to-end, and applied without per-signal parameter tuning.
- **Contribution 2**: Identification and quantification of the ℓ_1 leakage floor (BLI ≈ 0.59) as a *fundamental architectural limit* that loss engineering cannot overcome. This is a negative result with clear implications for the field.
- **Contribution 3**: The failure mode catalogue—CG gradient vanishing, non-learnable f_c, train/inference mismatch, softplus artefacts—provides practical guidance for anyone attempting algorithm unrolling.
- **Contribution 4**: Emergent parameter specialisation across layers, demonstrating that unrolled networks learn non-uniform iteration schedules that classical algorithms cannot express.
- **The framing**: this is a *methodological exploration* and *negative result study*, not a deployment-ready tool. The value is in *what we learned* about unrolling sparsity-based algorithms.

Thesis references: Sec. 1.3 (contributions), Sec. 5.7 (summary of findings).

Q55

Your advisor invented BEADS. How do we know this thesis is not just an exercise in making the advisor's algorithm look relevant? Where is the objectivity?

Answer Strategy

Key points:

- The thesis is arguably *harder* on BEADS than an independent researcher would be: it identifies three failure modes of BEADS (sparsity violation, filter bandwidth mismatch, spatially uniform regularisation), demonstrates that the ℓ_1 prior is fundamentally limited in dense-peak regimes, and shows that unrolling inherits these limitations.
- A thesis that just praised BEADS would be weaker. The negative findings (leakage floor, over-regularisation, CG gradient collapse) are honest scientific reporting.
- BEADS was chosen because it has the cleanest structure for unrolling (explicit MM iterations, scalar parameters, joint peak+baseline estimation). This is a technical motivation, not a promotional one.
- The thesis *motivates moving beyond BEADS* (group sparsity, U-HQ penalties) as the primary future direction.

Thesis references: Sec. 2.2.1.5 (BEADS failure modes), Sec. 5.1.2 (leakage analysis), Sec. 5.7 (honest summary).

Q56

Can you derive the soft-thresholding operator on the board right now? Show me that it is the proximal operator of the ℓ_1 norm, and explain why it causes shrinkage bias.

Answer Strategy

Whiteboard derivation (MUST be prepared):

1. The proximal operator: $\text{prox}_{\lambda\|\cdot\|_1}(\mathbf{v}) = \arg\min_{\mathbf{x}} \frac{1}{2}\|\mathbf{x} - \mathbf{v}\|^2 + \lambda\|\mathbf{x}\|_1$.
2. The problem separates element-wise: $\min_{x_i} \frac{1}{2}(x_i - v_i)^2 + \lambda|x_i|$.
3. Take subgradient, set to zero: $(x_i - v_i) + \lambda \cdot \text{sign}(x_i) = 0$.
4. Three cases: $v_i > \lambda \Rightarrow x_i = v_i - \lambda$; $v_i < -\lambda \Rightarrow x_i = v_i + \lambda$; $|v_i| \leq \lambda \Rightarrow x_i = 0$.
5. This is $\mathcal{S}_\lambda(v_i) = \text{sign}(v_i)\max(|v_i| - \lambda, 0)$.
6. **The bias**: for any non-zero true signal $x^* > 0$, $\mathcal{S}_\lambda(x^*) = x^* - \lambda < x^*$. The estimate is *always* shrunk by exactly λ. In dense-peak regions where many entries are non-zero, this shrinkage accumulates.

Thesis references: Eq. 3.14 (asymmetric soft-thresholding), Sec. 3.2.4 (proximal step).

Q57

Your masked baseline loss (Term 5) supervises the baseline only where peaks are absent. But at inference time, you do not know where peaks are. The model was trained with ground-truth peak masks that it will never have in deployment. Isn't the model learning to rely on information it cannot access?

Answer Strategy

Key points:

- The masked loss provides a *training signal*, not an inference-time input. The network does not receive the mask at test time—it receives only $\mathbf{y}$.
- The training objective teaches the network to produce baselines that are accurate in non-peak regions. At inference, the network generalises this learned behaviour to unseen signals.
- This is standard supervised learning: the ground truth is used to compute the loss during training but is not available at test time. All supervised losses "rely on information the model cannot access at inference."
- The concern would be valid if the mask were an *input* to the network. It is not—it only shapes the gradient.
- Potential issue: if the peak mask definition ($\gamma = 0.02$ threshold) is very different between training and real data, the learned baseline behaviour may not transfer.

Thesis references: Sec. 3.7.3 (masked baseline loss), Eqs. 3.27–3.29.

Q58

You claim "interpretability" as a key advantage. But can you actually diagnose *why LBEADS-NET fails on a specific signal by inspecting the learned parameters? Show me how.*

Answer Strategy

Key points:

- Partial yes: the per-layer parameters have physical meaning. If $\lambda_0^{(k)}$ is very large in early layers, one can diagnose "early layers are over-thresholding small peaks." If $r^{(k)}$ is low, "insufficient asymmetric penalisation of negative artefacts."
- More concretely: one can run the signal through each layer sequentially, inspecting the intermediate $\mathbf{x}^{(k)}$ and $\mathbf{f}^{(k)}$ at each stage. This "layer-by-layer decomposition" is impossible with a black-box U-Net.
- However, with K=20 layers and 5 parameters each, the interaction effects are complex. "Interpretable" does not mean "simple to diagnose."
- The emergent specialisation (Sec. 5.5) demonstrates that the parameters *are* interpretable post-training. But using them for per-signal failure diagnosis is an unsolved practical problem.
- Honest answer: "more interpretable than a black box" is correct, but "fully diagnosable" is an overstatement.

Thesis references: Sec. 5.5 (parameter analysis), Sec. 3.4.3 (layer chaining).

Q59

You exploit dense $O(N^2)$ matrices when the orignal BEADS uses $O(N)$ banded solvers. An undergraduate could implement banded matrix-vector products in PyTorch. Why didn't you?

Answer Strategy

Key points:

- Concede: banded matrix operations are straightforward in theory. The challenge is integrating them cleanly with PyTorch's autograd for backpropagation.
- The highpass filter involves $\mathbf{H} = \mathbf{B}\mathbf{A}^{-1}$, where $\mathbf{A}$ is banded. The forward pass requires solving $\mathbf{A}\mathbf{z} = \mathbf{v}$ (banded system solve). The *backward* pass requires differentiating through this solve, which PyTorch does not natively support for banded systems.
- Options: (a) use `torch.linalg.solve_triangular` with LU-factored $\mathbf{A}$, (b) write a custom autograd function for banded solves, (c) use implicit differentiation.
- All of these require more engineering effort than converting to dense, which was the pragmatic choice given the thesis timeline.
- This is the single most important engineering improvement for practical deployment.

Thesis references: Sec. 3.2.2 (filter construction), Sec. 3.4 (dense implementation).

Q60

ADMM-Net is the standard framework for unrolling constrained optimisation problems. Why did you not use ADMM? The baseline non-negativity constraint and the frequency separation are natural ADMM constraints.

Answer Strategy

Key points:

- ADMM decomposes the problem into sub-problems that handle different constraints/penalties separately, linked by dual variables (Lagrange multipliers). This would naturally separate peak sparsity, baseline smoothness, and frequency separation into distinct ADMM updates.
- BEADS is formulated as an unconstrained MM problem, not as a constrained optimisation. Converting to ADMM would require reformulating the cost function with explicit constraints, which changes the algorithm being unrolled.
- The thesis's goal is to unroll *BEADS specifically*, preserving its structure. ADMM-based unrolling would be a different method.
- ADMM introduces additional hyperparameters (penalty parameter ρ, dual variable update rate) that would also need to be learned per layer.
- Concede: ADMM unrolling is a promising alternative approach that should be explored.

Thesis references: Sec. 2.3.4 (ADMM-Net survey), Sec. 3.2 (ISTA choice rationale).

Q61

Your model is trained at N=4096 and tested at N=4096. Real chromatograms can have N=10^5 to 10^6 points. What happens when you apply this model to a different signal length? Does it even work?

Answer Strategy

Key points:

- **It does not work directly.** The dense matrices $\mathbf{A}$, $\mathbf{B}$, $\mathbf{L}$ are precomputed at construction time for a fixed N. A different signal length requires reconstructing these matrices (which changes their spectral properties) and re-registering the buffers.
- The 100 learned scalar parameters are *independent of N*—they can be transferred to any signal length. Only the filter matrices need to be recomputed.
- This is a strength of the architecture: the learned parameters are signal-length-agnostic. The filter construction adapts automatically via `BAfilt(d, fc, N_new)`.
- However, **this was not tested**. The normalised cutoff f_c=0.006 has a different physical meaning at different N (the period $1/f_c \approx 167$ samples is fixed, so longer signals have proportionally more baseline variation relative to the filter's passband).
- With banded implementation, arbitrary N would be trivial. With dense matrices, N=10^5 requires $\sim$80 GB per matrix—infeasible.

Thesis references: Sec. 3.4 (architecture), Sec. 3.2.2 (filter parameterisation).

Q62

You initialise all layers with identical parameters and let training differentiate them. But you only train for 30 epochs (300 gradient steps). With 100 parameters to differentiate in 300 steps, haven't the parameters barely moved from their initialisation? How much specialisation can actually emerge?

Answer Strategy

Key points:

- The 30-epoch / 300-step budget is indeed very tight for 100 parameters. This is why the more complex loss configurations (Exp. 1.5) fail—they need more steps.
- However, the progressive experiments show meaningful learning: Peak MSE drops from 21.27 (classical) to 9.30 (Exp. 1.3) in 300 steps. The parameters have clearly moved.
- The parameter analysis in Sec. 5.5 uses a different model (6 layers, 100 epochs = 1000 steps) where specialisation is more pronounced.
- Concede: the 30-epoch budget was a practical constraint (training time on CPU), not an optimised choice. Longer training would likely improve all configurations, especially the complex ones.
- This is precisely why the curriculum training (Sec. 3.6) exists: it provides a *structured path* through parameter space that achieves more in fewer steps than random initialisation.

Thesis references: Table 4.2 (training config), Sec. 5.5 (parameter analysis), Sec. 3.6 (curriculum).

Q63

Write the BEADS cost function on the board. Now take the gradient with respect to $\mathbf{x}$*. Now show me how one MM iteration relates to one ISTA step. Where does the approximation enter?*

Answer Strategy

Whiteboard derivation (MUST be prepared):

1. BEADS cost: $J(\mathbf{x}) = \frac{1}{2}\|H(\mathbf{y}-\mathbf{x})\|^2 + \lambda_0 \sum_n \theta(x_n) + \lambda_1 \sum_n \phi(\Delta x_n) + \lambda_2 \sum_n \phi(\Delta^2 x_n)$.
2. Split into smooth + non-smooth: $J = \underbrace{J_{\text{smooth}}}_{\text{data fidelity + smoothness}} + \underbrace{J_{\text{non-smooth}}}_{\lambda_0 \sum \theta(x_n)}$.
3. $\nabla J_{\text{smooth}} = -H^T H(\mathbf{y}-\mathbf{x}) + \lambda_1 D_1^T \psi(D_1\mathbf{x}) + \lambda_2 D_2^T \psi(D_2\mathbf{x})$ where $\psi(z) = z/(|z|+\epsilon)$.
4. **MM step**: majorise the non-quadratic terms, solve the resulting banded linear system exactly. This gives *exact* majoriser minimisation.
5. **ISTA step**: take one gradient step on J_{smooth}: $\mathbf{x}_{\text{update}} = \mathbf{x}^{k-1} + \eta \nabla J_{\text{smooth}}$, then apply proximal operator for θ: $\mathbf{x}^k = \mathcal{S}_{\text{asym}}(\mathbf{x}_{\text{update}}; \lambda_0 \eta, r)$.
6. **The approximation**: ISTA replaces exact MM minimisation with a single gradient step + proximal. It is less accurate per iteration but *much simpler* to differentiate.

Thesis references: Eq. 2.1 (BEADS cost), Algorithm 1 (MM), Algorithm 2 (ISTA layer), Sec. 3.3.

Q64

Your v5 model uses clamped exponential parameterisation, but Section 3.2.1 says "No clamping is applied—the exponential function alone guarantees positivity." Which is it? Does the 6-layer parameter analysis (Table 5.5) show clamped parameters (marked with ∗)? These statements contradict each other.

Answer Strategy

Key points:

- This is a genuine inconsistency in the thesis.
- Section 3.2.1 describes the v5 architecture (the final model) where parameters use unclamped exponential parameterisation: $\lambda = \exp(\theta)$, no bounds.
- Table 5.5 shows parameters from a *different model* (6-layer, N=1024, curriculum-trained) that uses clamped parameterisation (parameters marked with $*$ hit clamp boundaries).
- The two models have different parameterisation schemes. This is not made clear in the thesis and creates confusion.
- Fix: add a note in Sec. 5.5 that the 6-layer model uses a different (earlier) parameterisation with clamps, and clarify that the main 20-layer model uses unclamped exponential.

Thesis references: Sec. 3.2.1 (unclamped), Table 5.5 (clamped $*$ markers).

Q65

You claim the per-layer independence enables "emergent parameter specialisation." But with 100 parameters trained on 160 signals for 30 epochs, how do you distinguish "specialisation" from "overfitting to the training set"? What would the parameters look like on a different random seed?

Answer Strategy

Key points:

- Valid concern. The monotonic increase in λ_0 across layers could be a genuine pattern or a training artefact.
- To distinguish: train multiple models with different random seeds and check whether the same qualitative pattern (increasing sparsity, non-monotonic asymmetry, stable step size) appears. This was not done.
- Evidence against overfitting: the test-set metrics improve consistently across the progressive experiments, suggesting the learned parameters generalise. If the model were overfitting, we would expect test performance to diverge from training performance.
- The specialisation pattern is also *physically sensible*: it matches the behaviour of iterative thresholding algorithms (coarse-to-fine refinement). Overfitting would produce arbitrary or noisy parameter values.
- Concede: multi-seed experiments are needed to confirm the robustness of the specialisation pattern.

Thesis references: Sec. 5.5 (parameter analysis), Table 5.5 (learned values).

12 Known Inconsistencies & Issues to Fix Before Defense

CRITICAL: Fix Before Defense

The following issues have been identified in the current thesis draft. Each should be resolved before the defense to avoid uncomfortable questions.

#	Issue	Location	Fix
1	Abstract is Lorem Ipsum placeholder text	Abstract page	Write the real abstract
2	Conclusions chapter is Lorem Ipsum placeholder	Chapter 6	Write real conclusions
3	Dedication has "YOUR DEDICATION (1-2 LINES)" placeholder	Dedication page	Write dedication or remove
4	Broken cross-references showing "Chapter ??"	pp. 43, 56, 68	Fix `\label`/`\ref` pairs
5	University ID and NetID are placeholders ("N#", "NetID")	Title/signature pages	Insert real ID numbers
6	Ablation tables 5.3 and 5.4 have all dashes	Tables 5.3–5.4	Run experiments or remove
7	Sections 5.3 (Qualitative Examples) and 5.4 (Training Dynamics) are placeholder	Sec. 5.3–5.4	Write or mark as future work
8	Advisor title inconsistency: "Assistant Prof." vs "Professor"	Abstract vs. approval page	Verify correct title, unify
9	Loss term count: "11-term" in Intro vs "twelve" in Sec. 3.7	Sec. 1.2 vs Sec. 3.7	Correct Intro to "twelve-term"
10	Peak model mismatch: EMG in Sec. 2.1.4 vs bi-Gaussian in Sec. 4.1.1	Sec. 2.1.4 vs 4.1.1	Add justification in Sec. 4.1.1

Table 1: Known issues in the current thesis draft, ordered by severity.

Priority Actions

P0 Items 1, 2, 5: These will be immediately noticed by any committee member opening the document.

P1 Items 3, 4, 8, 9: Visible quality issues that suggest the document was not carefully reviewed.

P2 Items 6, 7, 10: Content gaps that may or may not be raised depending on committee member interests.

General Defense Advice

Guiding Principles for the Defense

1. **Own the negative results.** The BLI plateau at 0.59 and the over-regularisation finding (Exp. 1.5) are *contributions*, not failures. Present them as deliberate investigations. "We tried X, and it did not work because of Y" is a perfectly valid scientific finding.

2. **Know what you do not know.** When a question targets a gap (e.g., "why no arPLS comparison?"), say: "That is a limitation of this work. Here is what I would do if I had more time." Do not bluff.

3. **Redirect to the architecture.** Many attack questions (area error, BLI, peak correlation) have the same root cause: ℓ_1 shrinkage bias. Learn the shrinkage bias argument cold and redirect to it.

4. **Prepare the whiteboard.** Be ready to derive: (a) the BEADS cost function (Eq. 2.24), (b) the ISTA proximal step (Eq. 3.9), (c) the soft-thresholding operator and its bias, (d) why untied parameters break MM convergence.

5. **Credit the advisor.** Professor Selesnick co-authored BEADS. When discussing the algorithm, frame the work as a natural evolution of the group's research programme.

6. **Time management.** Aim for 25–30 minutes of presentation, leaving 30–45 minutes for questions. For each question, answer in 60–90 seconds. If the committee wants more detail, they will ask.

7. **Do not read from this document during the defense.** Internalise the key points. The goal is fluency, not recitation.

Prepared April 2026.
Good luck, Saad.

Glossary of Key Terms

Terms Defined in the Context of This Thesis

This glossary defines terms as they are used specifically within the LBEADS-NET thesis. Definitions are scoped to the chromatographic baseline-correction context and may differ from their broader usage in other fields.

ℓ_1 Leakage Limit

The empirically observed floor on the Baseline Leakage Index (BLI $\approx$ 0.59) that persists across *all* LBEADS-NET loss configurations (Experiments 1.1–1.5). The limit arises because the ℓ_1-based soft-thresholding operator at the core of every ISTA layer shrinks *every* non-zero signal value by exactly λ (the effective threshold). In dense-peak regions where the sparsity assumption is violated, this shrinkage accumulates: the total missing peak energy is proportional to $\lambda \times$ (number of non-zero samples), and this energy is absorbed into the baseline estimate via the low-pass filter. Because the shrinkage is a structural property of the proximal operator—not a tunable weight—no amount of loss engineering (anti-leakage penalties, asymmetric baseline loss, envelope constraints, etc.) can push the BLI below this floor. The limit generalises to *any* unrolled network that inherits ℓ_1-based sparsity priors, as independently confirmed by Gharbi et al. (2024) in their work on unrolled sparse-recovery architectures. *Thesis refs: Sec. 5.1.2, Table 5.2, Sec. 1.3 (Contribution 3)*